

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Information Technology In-Semester Examination-II Academic Year: 2025-26 (Semester - I)	
Class: B. Tech (IT)	Subject: Soft Computing	Subject Code: PEC-IT410
Date: 14.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note: Solve any four.

Draw neat diagrams, flow chart wherever necessary.

Q.1	Describe how Genetic Algorithms can be applied to feature selection in machine learning.	5	CO3	BT3
Q.2	What is supervised learning in neural networks? Give an example.	5	CO2	BT2
Q.3	Draw and explain the block diagram of ANN. Differentiate between Artificial Neural Network (ANN) and Biological Neural Network(BNN).	5	CO2	BT1
Q.4	Explain how adaptive resonance architecture solves the stability-plasticity dilemma?	5	CO2	BT5
Q.5	How does GA compare with gradient descent in optimization problems?	5	CO3	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Students will be introduced to Fuzzy logic and its applications.
CO 2	Student should have gain knowledge of artificial neural networks and its applications.
CO 3	Student have to solve single-objective optimization problems using GAs.
CO 4	Students should understand solving multi-objective optimization problems using Evolutionary algorithms (MOEAs).
CO 5	Students will understand the basic concepts of Applications of Soft computing to solve problems in varieties of application domains.

	S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded. ISE 2 Examination - Semester I Academic Year – 2025 - 26 Class: Final Year (Information Technology)		
Date: 14/10/2025 Time: 14:30 to 15:15	Subject: Elec – Human Computer Interaction	PEC – IT409	Max. Marks: 20

Notes:

1. SOLVE ANY FOUR QUESTIONS.
2. This paper contains 1 page.
3. Figures to right indicate full marks.
4. Assume suitable data if required and mention clearly.

Q. No.	Question	Marks	CO/BT
Q. 1	'What Is Design'? State the golden rule of design. Explain the Interaction design process.	[05]	CO2/BT 2,3
Q. 2	State and explain the 8 golden rules given by Shneiderman.	[05]	CO2/BT 1,2
Q. 3	Write a short note on various things to be considered during navigation design and screen design of an interactive system.	[05]	CO3/BT 2,3
Q. 4	Explain in detail the activities in the modified Waterfall model of the software life cycle related to UI development in reference to HCI.	[05]	CO4/BT 4, 5
Q. 5	State and explain Norman's 7 principles of design.	[05]	CO2/BT 1,2

//ALL THE BEST//

Course Outcomes (CO):

- CO1. To introduce the concept of human-computer-interaction in study.
- CO2. To gain the knowledge of Interaction design methodologies.
- CO3. To design and evaluate effective human-computer-interactions.
- CO4. To study HCI models and theories.
- CO5. To understand HCI design processes.
- CO6. To introduce how to apply the Human-Computer Interaction concepts to the current interaction designs.

Bloom's Taxonomy (BT):

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